POLITECNICO DI MILANO

1) Computing Infrastructures

Prof. Manuel Roveri

Why?

2023-2024 …

Nassim Taleb “The Black Swan” (2007)

2023-2024 …

«History and societies do not crawl, they leap: moving from one fracture to another with some vibrations in between.»

Nassim Taleb “The Black Swan” (2007)

The Black Swan

When?

Where?

How?

The Black Swan

Unexpected and unpredictable

When?

Significant effects

Where?

Latent precursors

How?

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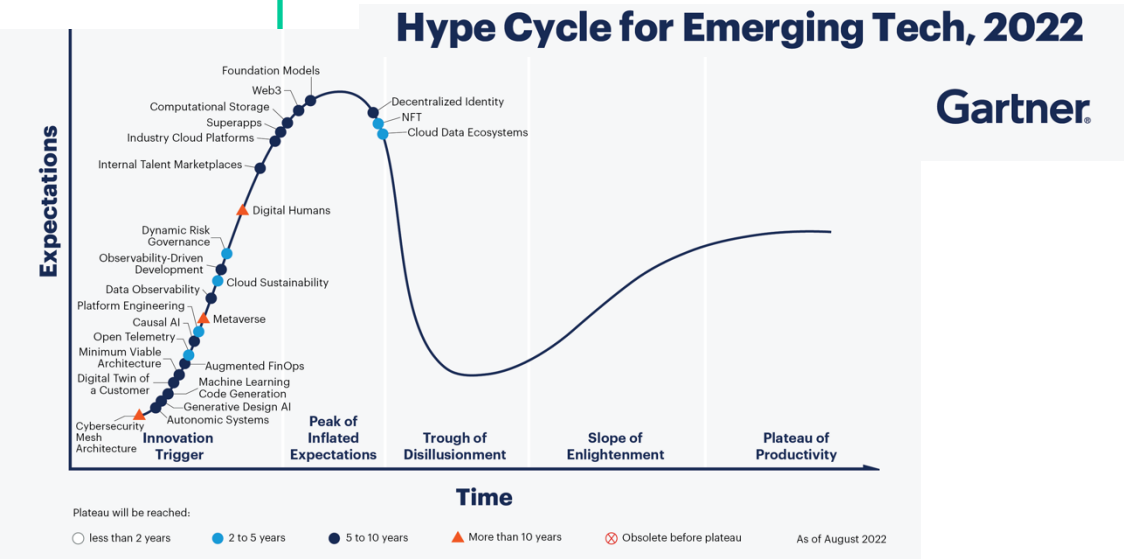

The Black Swan

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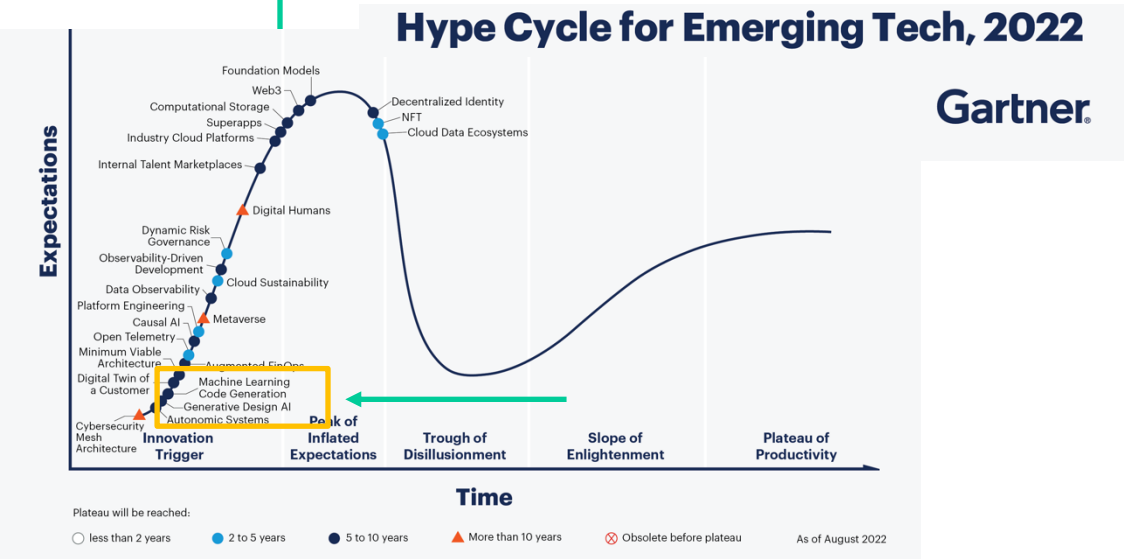

The Black Swan

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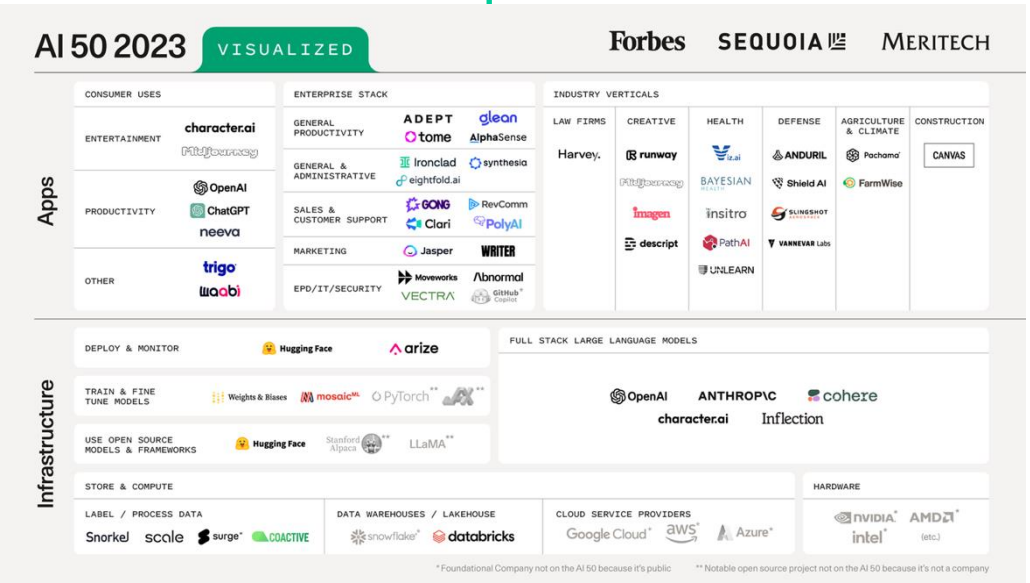

Source, Sequoia Capital Report 2023, Generative AI

The Black Swan

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The Guardian, 08/09/2020

The Black Swan

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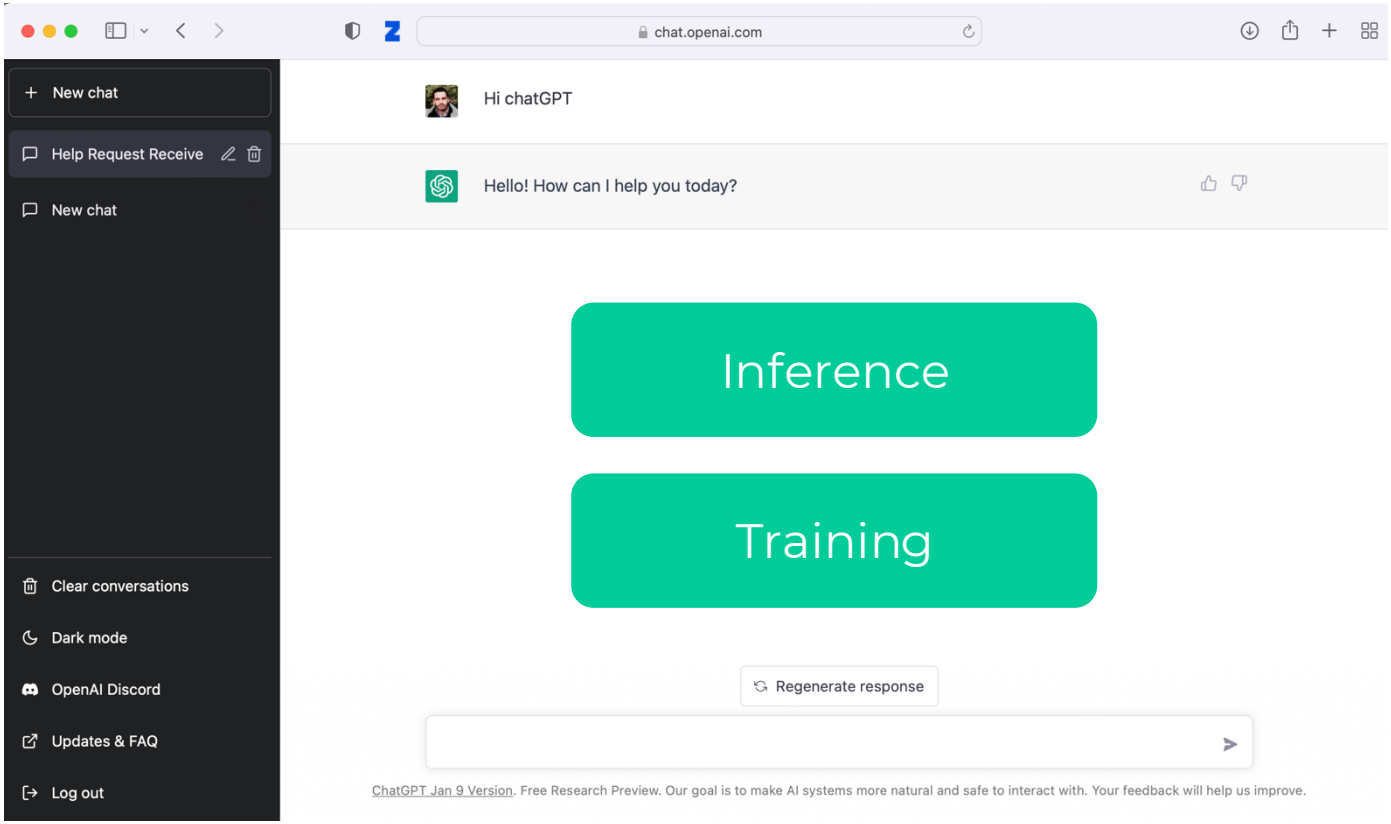
The most important question … why?



The topics of the course: what are we going to see today?

A. HW Infrastructures:
- **System-level**: Computing Infrastructures and Data Center Architectures, Rack/Structure;
- **Node-level**: Server (computation, HW accelerators), Storage (Type, technology), Networking (architecture and technology);
- **Building-level**: Cooling systems, power supply, failure recovery

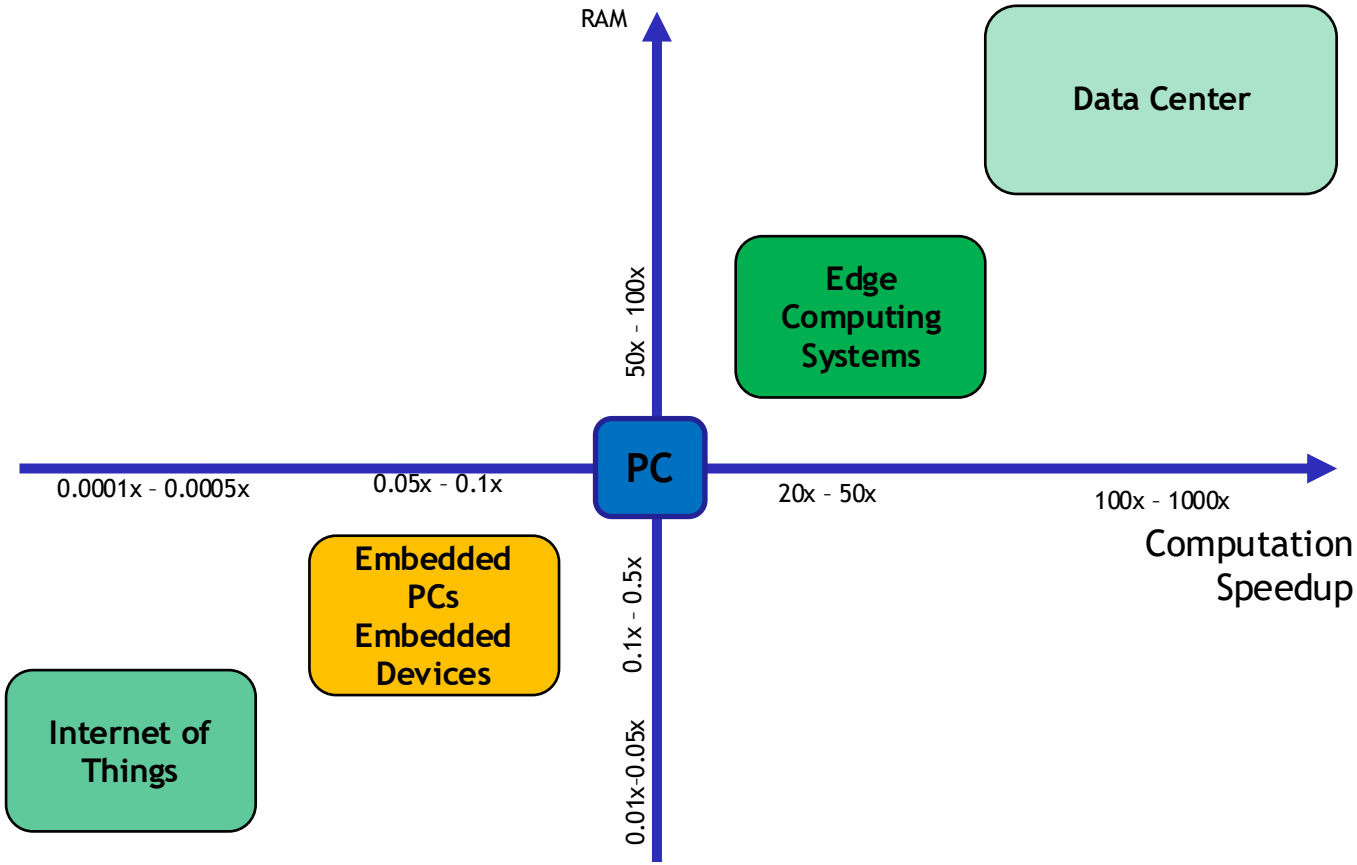
B. SW Infrastructures:
- **Virtualization**: Process/System VM, Virtualization Mechanisms (Hypervisor, Para/Full virtualization)
- **Computing Architectures**: Cloud Computing (types, characteristics), Edge/Fog Computing, X-as-a service
- **Machine and deep learning-as-a-service**

C. Methods:
- **Reliability and availability of datacenters** (definition, fundamental laws, RBDs)
- **Disk performance** (Type, Performance, RAID)
- **Scalability and performance of datacenters** (definitions, fundamental laws, queuing network theory)

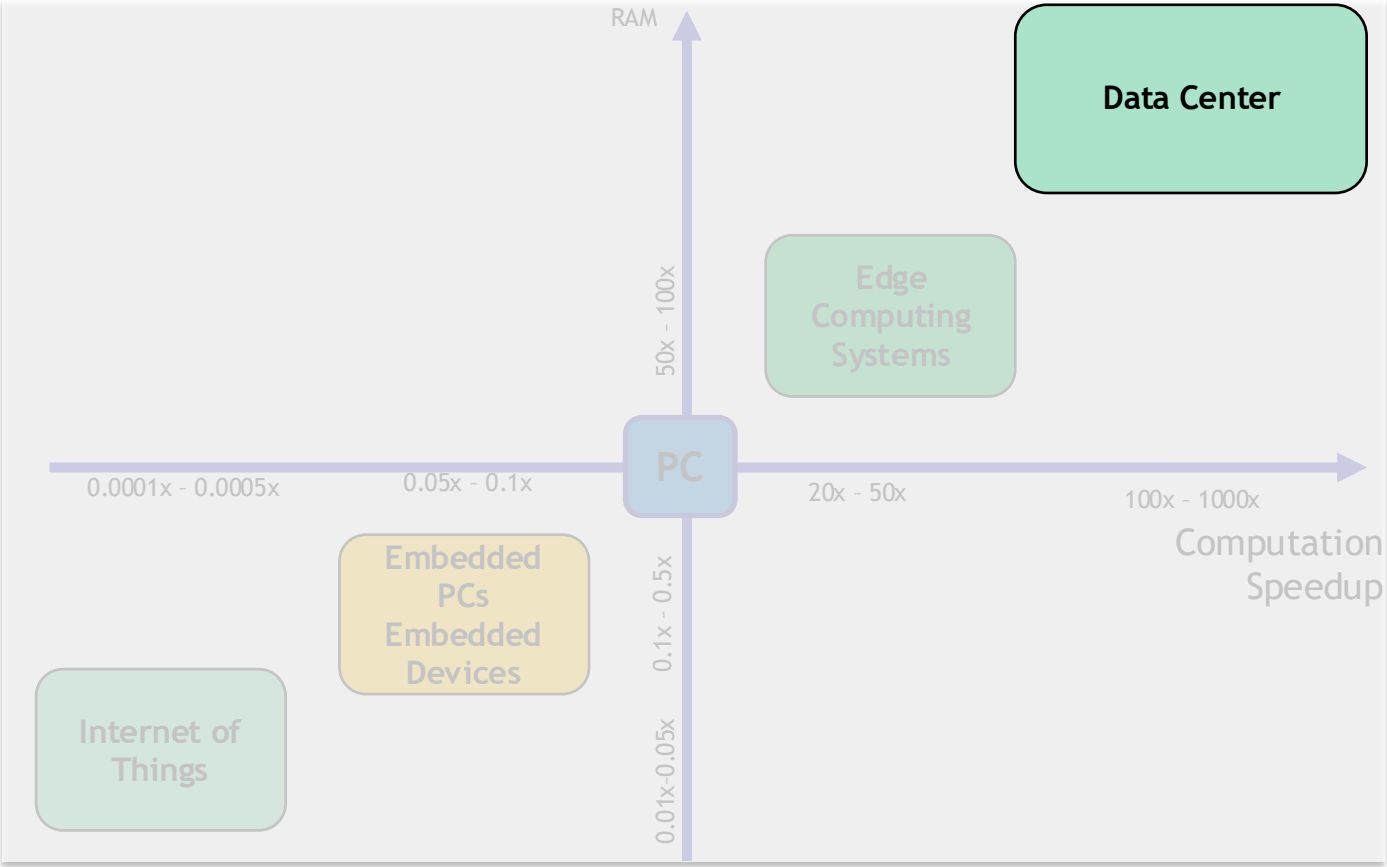
What is a computing infrastructure?

Technological infrastructure that provides hardware and software for computation to other systems and services

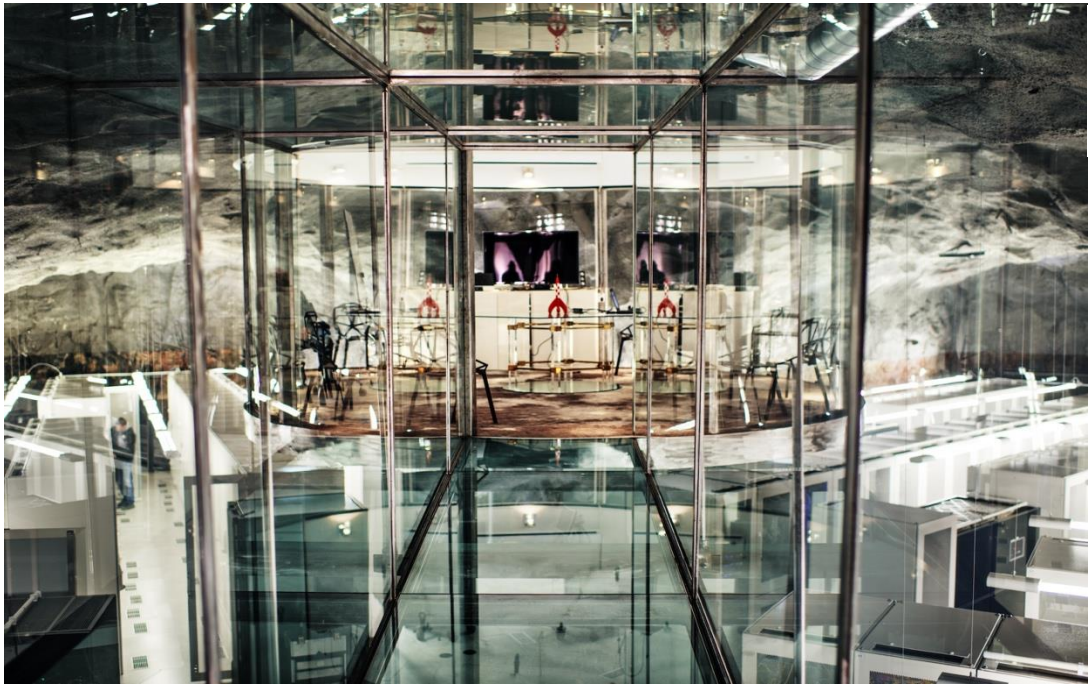
Examples of Computing Infrastructures



Examples of Computing Infrastructures



Data Centers: a technological perspective



The Pionen White Mountains is a Swedish data center. This center is located in Stockholm and is one of the largest data centers in the world.

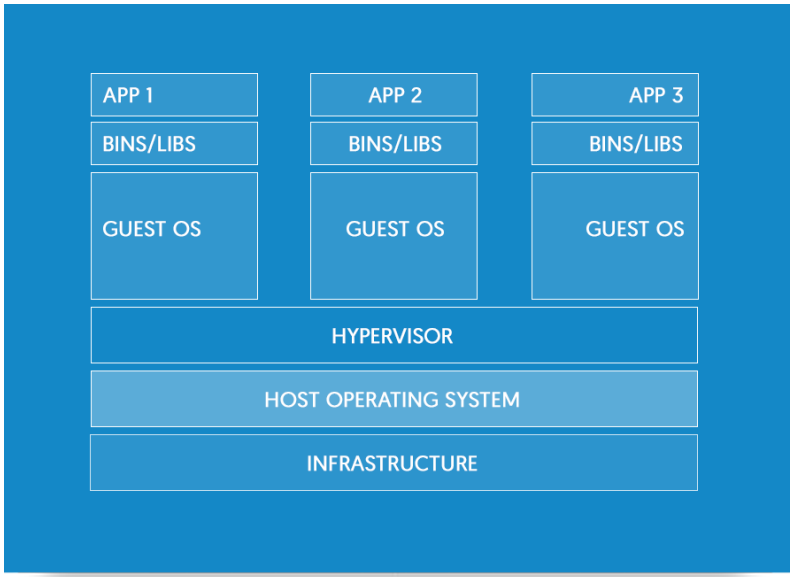
Server for Processing

Server for Storage

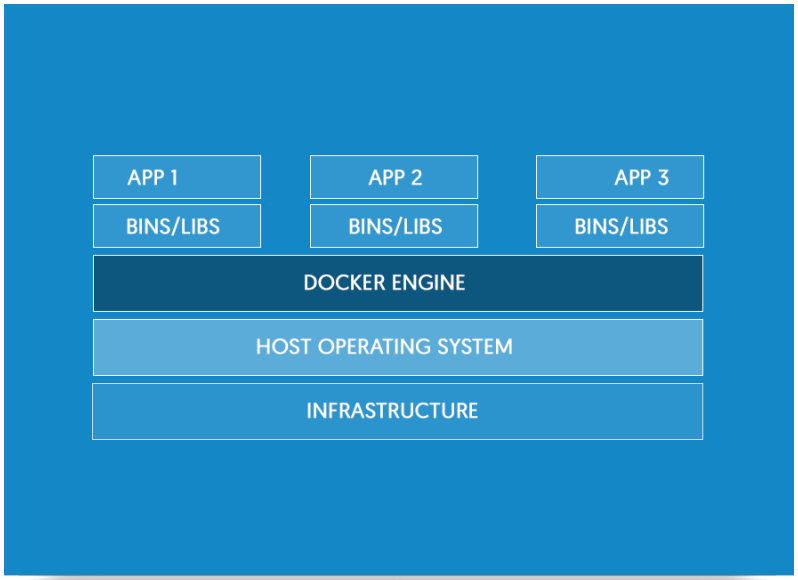
Server for Communication

Virtual Machines and Containers

VMMs
Provide the full stack (OS, LIB, APP).
Applications depend on guest OS.



Containers
Applications are packaged with all their dependencies into a standardized unit for software development/deployment.



Data Centers: advantages and disadvantages

- ✓ Lower IT costs
- ✓ High performance
- ✓ Instant software updates
- ✓ "Unlimited" storage capacity
- ✓ Increased data reliability
- ✓ Universal document access
- ✓ Device Independence

- ➢ Require a constant Internet connection
- ➢ Do not work well with low-speed connections
- ➢ Hardware Features might be limited
- ➢ Privacy and security issues
- ➢ High Power Consumption
- ➢ <u>Latency in making decision</u>

Data Centers: advantages and disadvantages

WATER

Waterlogged

A midsize data center uses roughly as much water as about 100 acres of almond trees or three average hospitals, and more than two 18-hole golf courses.

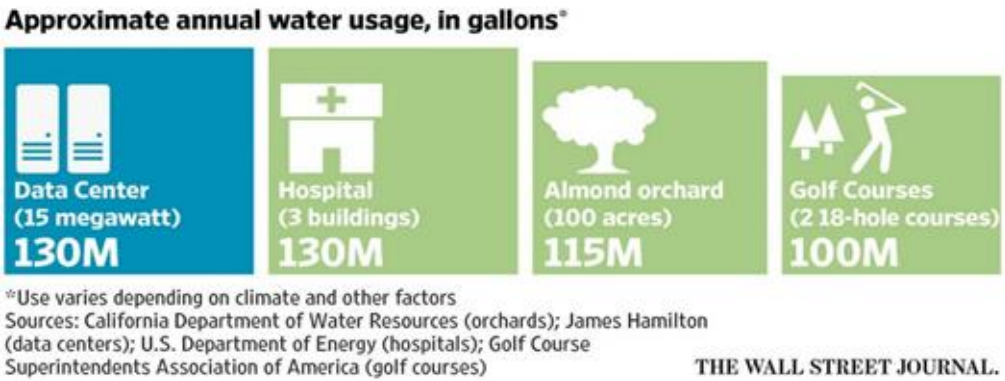
Approximate annual water usage, in gallons*

*Use varies depending on climate and other factors
Sources: California Department of Water Resources (orchards); James Hamilton (data centers); U.S. Department of Energy (hospitals); Golf Course Superintendents Association of America (golf courses)

THE WALL STREET JOURNAL.

Data Centers: advantages and disadvantages

WATER

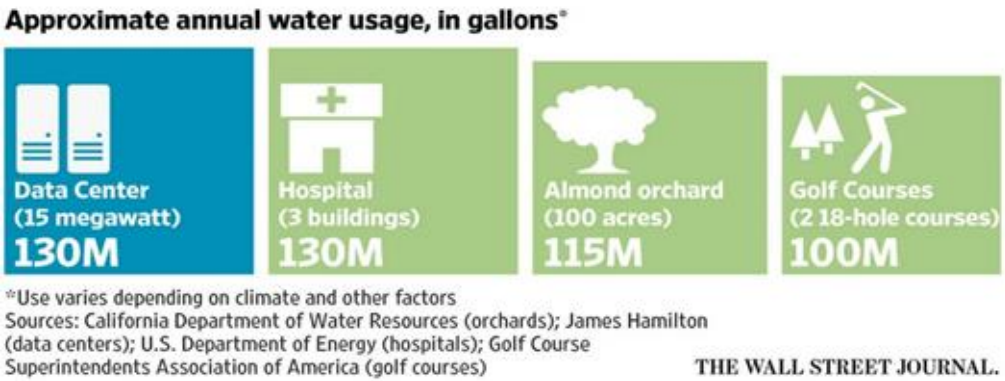


1%

Overall worldwide total energy consumption due to datacenters

Data Centers: advantages and disadvantages

WATER

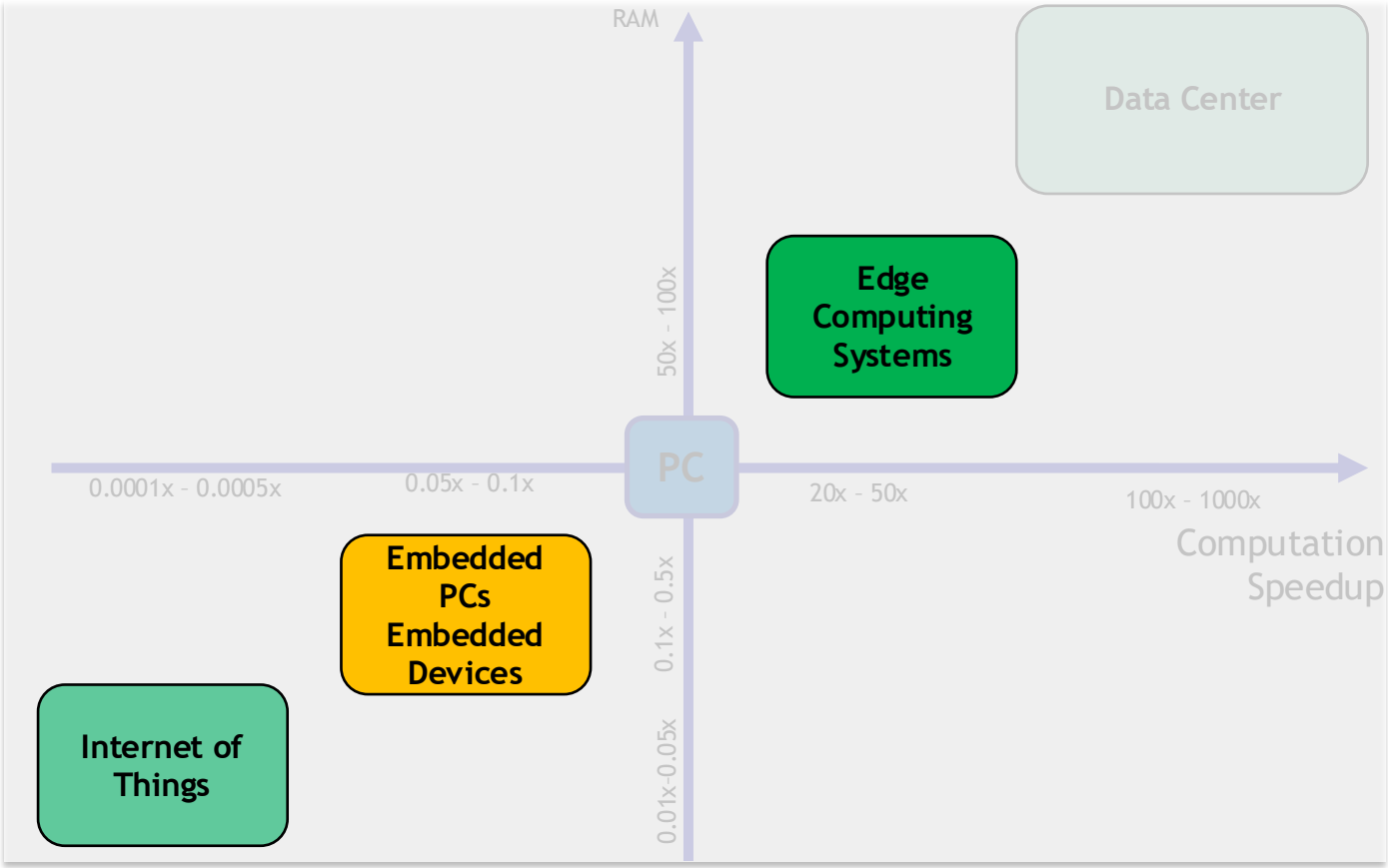


1%

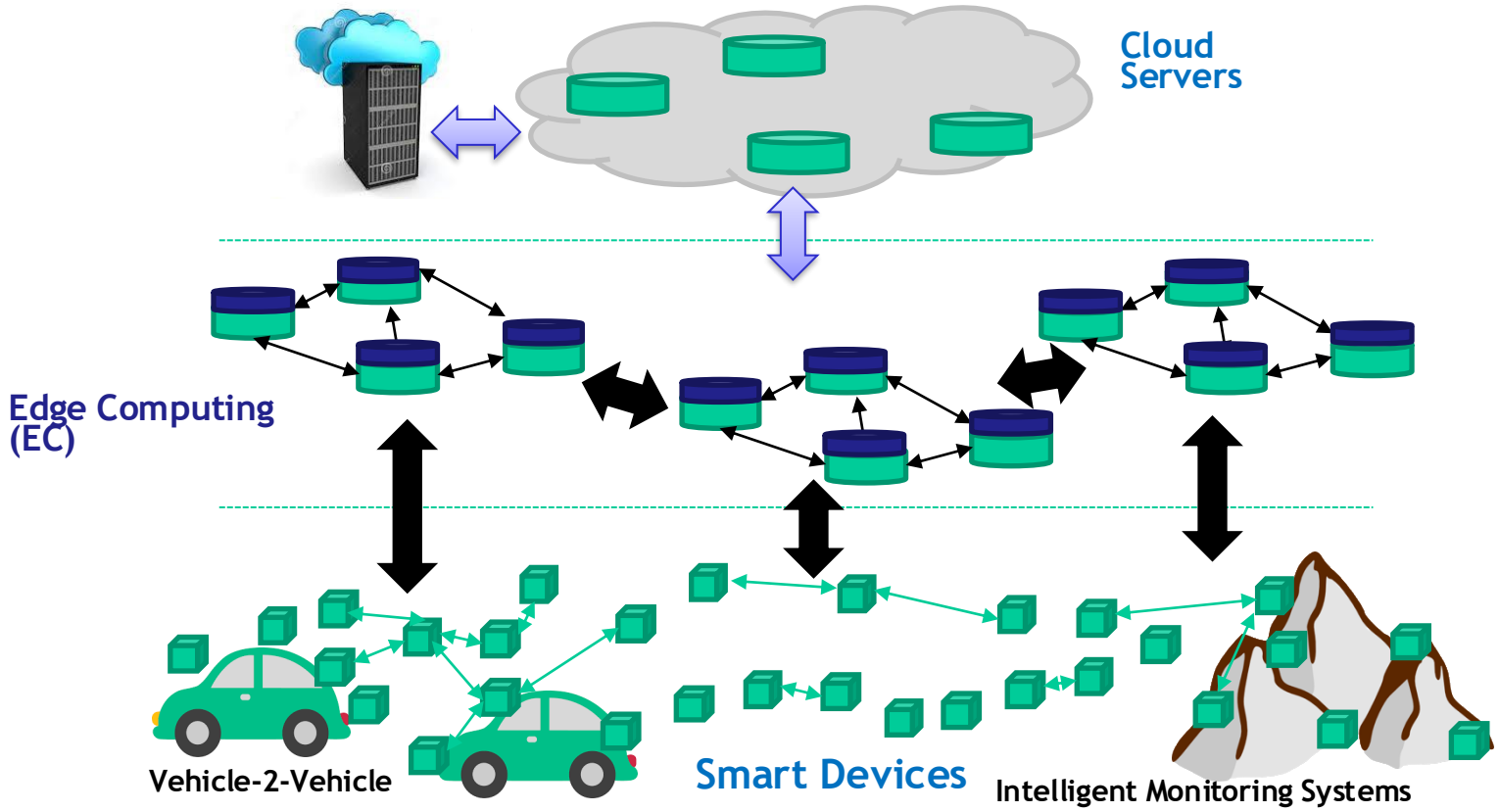
Overall worldwide total energy consumption due to datacenters

Amortized Cost	Component	Sub-Components
~45%	Servers	CPU, memory, disk
~25%	Infrastructure	UPS, cooling, power distribution
~15%	Power draw	Electrical utility costs
~15%	Network	Switches, links, transit

Edge Computing, PC Embedded and IoT



Edge/Fog Computing Systems



Edge/Fog Computing Systems

MODELS	WORKLOADS	RACK UNITS	PROCESSOR	MAX MEMORY	MAX STORAGE	GPU
PowerEdge XE9680 Open 3D viewer	AI and machine learning, DL large data set training, HPC, CRISP, and healthcare	6U	2 x 5th Generation Intel® Xeon® Scalable processors	32 x 128 GB DDR5 (4 TB)	16 x E3.S or 8 x 2.5" SAS/SATA/NVMe (122.88 TB)	8 x (NVIDIA A100, H100, H200, or AMD MI300X)
PowerEdge XE9640 Open 3D viewer	AI, machine learning, DL training modeling and simulation, HPC, and healthcare	2U	2 x 5th Generation Intel® Xeon® Scalable processors	32 x 128 GB DDR5 (4 TB)	4 x 2.5" NVMe (61.44 TB)	4 x NVIDIA H100
PowerEdge XE8640 Open 3D viewer	AI, machine learning, DL medium data set training,HPC, finance, and research	4U	2 x 5th Generation Intel® Xeon® Scalable processors	32 x 128 GB DDR5 (4 TB)	8 x E3.S or 8 x 2.5" SAS/SATA/NVMe (122.88 TB)	4 x NVIDIA H100
PowerEdge R760xa Open 3D viewer	AI, machine learning, DL training and inferencing, HPC, and render farms and virtualisation	2U	2 x 5th Generation Intel® Xeon® Scalable processors	32 x 256 GB DDR5 (8 TB)	8 x 2.5" SAS/SATA/NVMe (122.88 TB)	4 x 400W (DW) or 12 x 75W (SW)

- ✓ High computational capacity
- ✓ Distributed computing
- ✓ Privacy and security
- ✓ Reduced Latency in making a decision

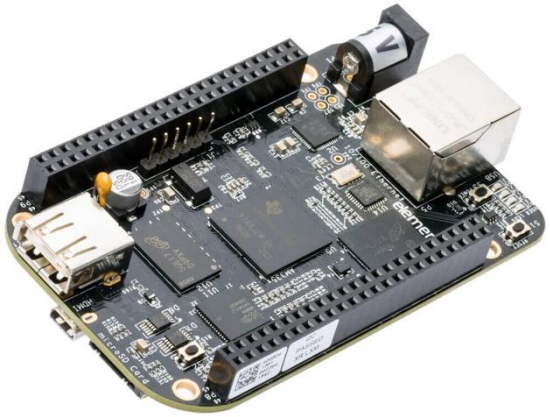
- ➢ Require a power connection
- ➢ Require connection with the Cloud

Embedded PCs

Raspberry Pi 5 8GB Quad-Core ARMA76 (64 Bits - 2,4 GHz)

Nvidia Jetson Orin™ Nano 8 GB 6 x 1.5 GHz

- ✓ Pervasive computing
- ✓ High performance unit
- ✓ Availability of development boards
- ✓ Programmed as PCs
- ✓ Large community



BeagleBone Black - CPU ARM Cortex A8 1GHz - 512MB RAM

Google Coral Dev Board, 1 GB RAM

- ➢ Pretty high power consumption
- ➢ (Some) HW design has to be done

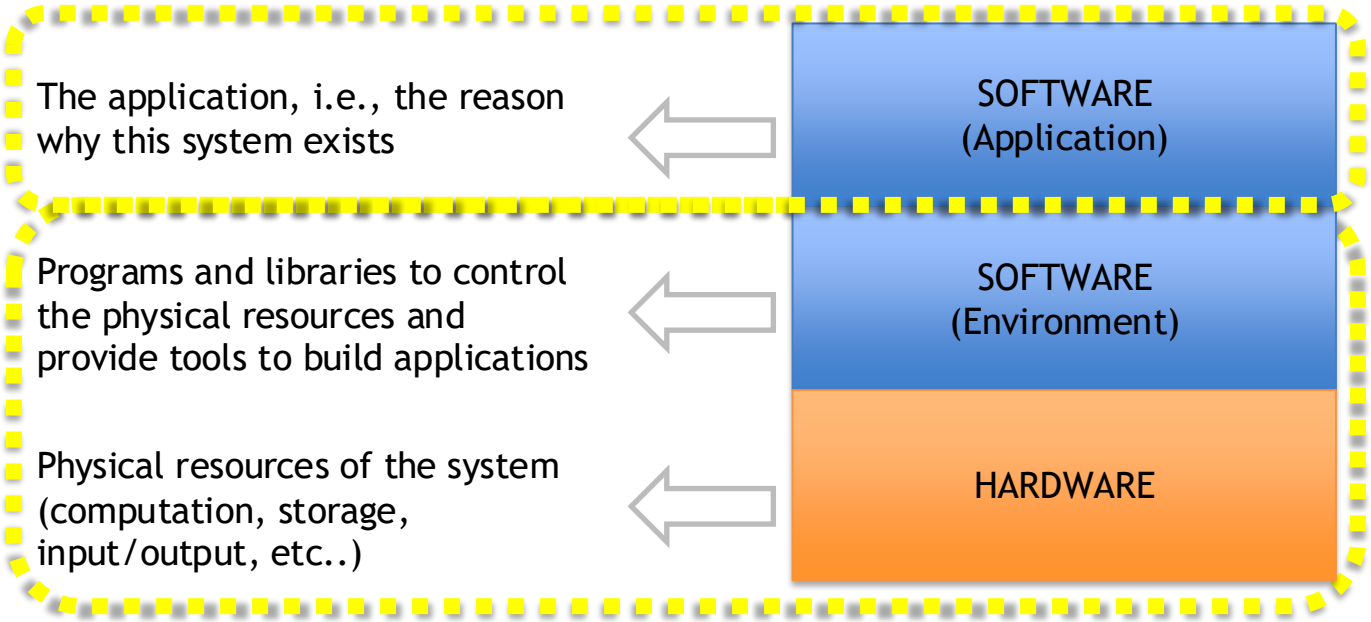
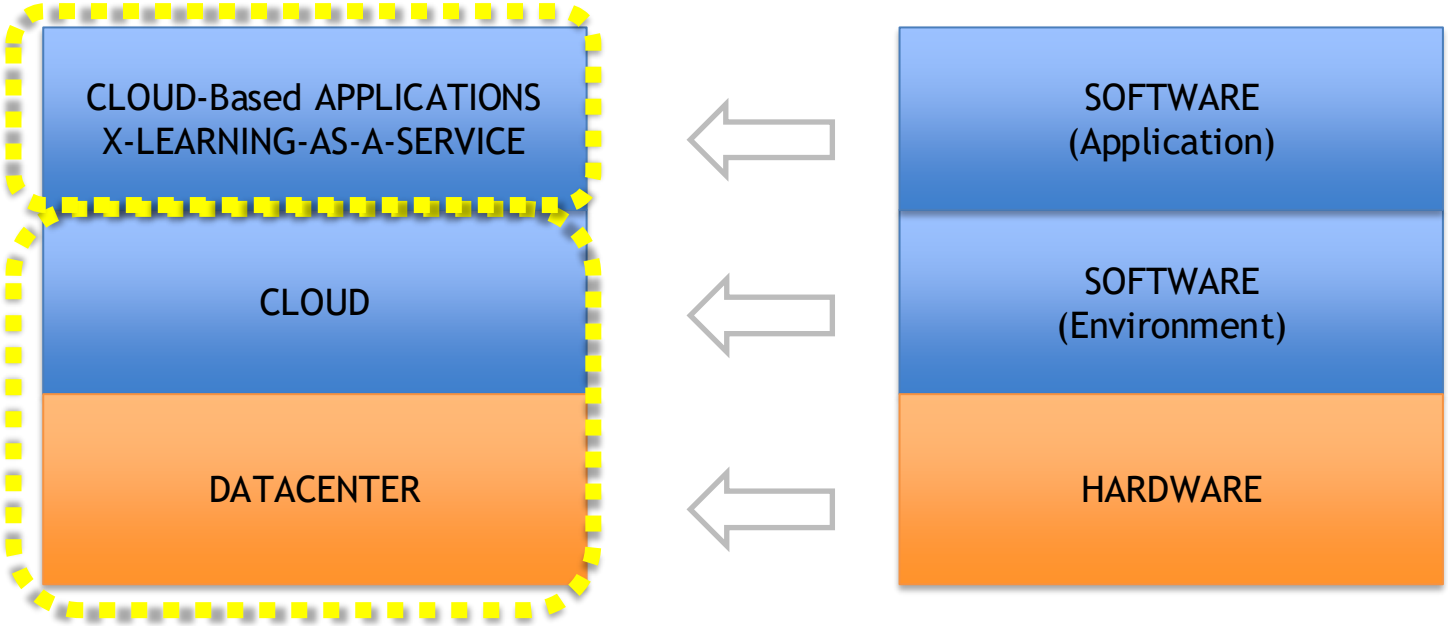
Internet-of-Things

	Arduino Nano	Nano ESP32-S3	Nano 33 IoT	Nano 33 BLE	Nano 33 BLE Sense Rev2	Nano RP2040 Connect	Nano Every	Nano Matter
Microcontroller	ATmega328	ESP32-S3	SAMD21 Arm® Cortex®-M0+ / u-blox® NINA-W102	nRF52840 / u-blox® NINA-B306	nRF5284 / u-blox® NINA-B306	Raspberry Pi RP2040 / u-blox® NINA-W102	ATMega4809	Silicon Labs MGM24
USB connector	Mini-B USB	USB-C®	Micro USB	Micro USB	Micro USB	Micro USB	Micro USB	USB-C®
I/O voltage	5 V	3.3 V	3.3 V	3.3 V	3.3 V	3.3 V	5 V	3.3 V
Input range	7-12 V	5-21 V	7-12 V	7-12 V	7-12 V	5-21 V	7-12 V	5-21 V
Clock speed	16 MHz	up to 240 MHz	48 MHz	64 MHz	64 MHz	133 MHz	16 MHz	78 MHz
SRAM	2 kB	512 kB	256 kB	256 kB	256 kB	264 kB	6 kB	256 kB
Flash	32 kB	16 MB	1 MB	1 MB	1 MB	16 MB	48 kB	1536 kB

- ✓ Highly Pervasive
- ✓ Wireless connection
- ✓ Battery Powered
- ✓ Low costs
- ✓ Sensing and actuating

- ➢ Low computing ability
- ➢ Constraints on energy
- ➢ Constraints on memory (RAM/FLASH)
- ➢ Difficulties in programming

An IT perspective for Computing Infrastructures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